



Molecular assessment of carbapenem-resistant and ESBL *Klebsiella pneumoniae* clinical isolates to decipher the correlation of antimicrobial resistance with virulence traits

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ABSTRACT

Klebsiella pneumoniae is a nosocomial pathogen that poses a serious concern due to the high prevalence of extended-spectrum beta-lactamases (ESBL) and carbapenem-resistant *K. pneumoniae* (CRKP) strains. However, the data on the prevalence of contributing virulence determinants are limited in the Pakistani population. The study aims to characterize clinical isolates of *K. pneumoniae* to understand clonal relationships and determine the relationship of antibiotic resistance with different virulence factors (i.e., biofilm, capsular polysaccharide, hemolysis, efflux pump, and outer membrane porins). The clinical strains were collected from the diagnostic facility of two public sector hospitals in Karachi. The resistance and virulence profile of the isolates were evaluated via antibiotic susceptibility test, double disk synergy test (DDST), ChromAgar, string test, blood hemolysis, and biofilm assay. Genotypically, the isolates were identified by 16S rRNA and rpoB gene, and further characterized for the presence of ESBL, CRKP, biofilm, efflux pump, and outer membrane porins genes. The clonal lineage among isolates was established by Enterobacterial Repetitive Intergenic Consensus Polymerase Chain Reaction (ERIC PCR). The antibiotic susceptibility test was analyzed through the Multiple Antibiotic Resistance (MAR) index, which revealed 90.2 % ($n = 102$) strains were MDR. Whereas, the genetic diversity was revealed through ERIC PCR and clade wise data revealed genetic variations due to ESBL (85 %), carbapenem resistance (73 %), biofilm (97 %), efflux pump (40–53 %), outer membrane porins (38–49 %), and hypermucoidity (6 %). The Pearson correlation analysis revealed a strong relationship of MDR strains with biofilm ($r = 0.99$), efflux pump ($r = 0.92$), and outer membrane porins ($r = 0.88$). The study highlighted the prevalence of MDR *K. pneumoniae* with the plethora of virulence factors in the local clinical setting, necessitating stringent screening to develop national policies to tackle further antimicrobial resistance development and outbreaks.

1. Introduction

Klebsiella pneumoniae is an opportunistic pathogen, clinically relevant in causing hospital or community-acquired infections such as pneumonia, urinary tract infection, bacteremia, and liver cirrhosis

(Abbas et al., 2024; Mirzaie and Ranjbar, 2021). *K. pneumoniae* has an open pan-genome and large accessory genome, acquiring antibiotic resistance genes (beta lactamases and carbapenemase) and virulence determinants (capsule formation, biofilm formation, iron uptake system, siderophores, etc.) that enhance the pathogenicity (Karampatakis et al.,

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2023). Due to their large genome and metabolic versatility, they can colonize various clinical or environmental niches. The major factor contributing to emerging antimicrobial resistance is the misuse of antibiotics in clinical settings or livestock as growth promoters (Aslam et al., 2022). The rampant usage of antibiotics by populations of developing countries contributes to antimicrobial resistance (AMR), hence the emergence of multidrug-resistant strains causing nosocomial infections (Jain et al., 2021; Taha et al., 2023). Pakistan is also among the low- and middle-income countries where AMR rates are high due to self-medication, misuse and overuse of antibiotics, and unnecessary recommendation of antibiotics for viral infections without culture and sensitivity testing (Bilal et al., 2021; Furqan et al., 2022; Saleem et al., 2023).

Besides antimicrobial resistance, the virulence factors such as capsular polysaccharides, efflux pumps, outer membrane porin, and urease accessory proteins also contribute to the severity of these notorious infections in clinical settings (Guerra et al., 2022; Mirzaie and Ranjbar, 2021). The transmission of antibiotic resistance and virulence factors also occurs through mobile genetic elements such as plasmids (Popa et al., 2024). The plasmid-encoded genes such as *bla_{CTX}*, *bla_{SHV}*, *bla_{TEM}*, *bla_{NDM}*, and *bla_{OXA-48}* *bla_{KPC}*, have been found and are responsible for beta-lactamases and carbapenemase resistance in *K. pneumoniae* strains (Hosdurg Bhaskar et al., 2019; Hussein et al., 2023). Carbapenem resistant *K. pneumoniae* (CRKP) are rapidly proliferate in clinical setting and causing high mortality. Some studies also revealed that loss of outer membrane porins and efflux pump system play an important role in drug ineffectiveness against the commonly used antibiotics extended spectrum beta lactam (ESBL) and carbapenem drugs (Hussein et al., 2024; Sanikhani et al., 2024). Biofilm formation is another virulence displayed by *K. pneumoniae* in a clinical setup. It is an impervious extracellular matrix which provides resistance to antimicrobials, chemicals, and the host immune system to its residents thus, facilitating persistent infection (Zehra et al., 2025). More than 80 % of microbial infections involve biofilm formation, which acts as a hinderance in selecting treatment options (Asghar et al., 2022). Among *Klebsiella pneumoniae*, factors which are involved in the biofilm formation are capsule, fimbriae, and pili. The most dominant fimbriae genes (*FimH*, *MrkA* *MrkD*) are associated with MDR clinical isolates of *K. pneumoniae* (Guerra et al., 2022; Mirzaie and Ranjbar, 2021). There is a paucity of knowledge on the prevalence of *K. pneumoniae* linked to antibiogram and correlation with virulence determinants such as biofilm in the population. Thus, molecular profiling of the bacterial strains is an important tool to link virulence factors and antimicrobial-resistant genes. Furthermore, it will also evaluate the genetic relatedness of nosocomial infections caused by *K. pneumoniae*. Genotyping by Enterobacterial Repetitive Intergenic Consensus Polymerase Chain Reaction (ERIC-PCR) is a major tool to investigate predominant strains involved in nosocomial infection. The clonal relatedness among *K. pneumoniae* strains can be analyzed by ERIC sequences to form clusters, which will be used to explore diversity (Bobbadi et al., 2023; Kundu et al., 2022).

The ongoing global convergence of the multidrug-resistant (MDR) strain of *K. pneumoniae*, especially extended-spectrum beta-lactamases, carbapenemase strains with other virulent determinants of *K. pneumoniae*, indicates an alarming situation, which deserves detection and monitoring to prevent the further spread (Kumar et al., 2023). Thus, this study was conducted for phenotypic and genotypic characterization of extended-spectrum beta-lactamases and carbapenem-resistant *K. pneumoniae* isolates and their correlation with virulence traits.

2. Material and methods

2.1. Collection and identification of strains

Klebsiella pneumoniae local clinical isolates were collected between December 2021 and December 2023 from different diagnostic units of

two public sector hospitals of Karachi, where the outpatient department handles more than 1000 cases every day due to free or very nominal charges. The strains were re-identified as *Klebsiella pneumoniae* through Gram staining, streaking on MacConkey agar, biochemical identification (Remel RapID™ ONE SYSTEM, Thermoscientific, USA), and presence of *K. pneumoniae* specific *16SrRNA*, and *rpoB* genes, ATCC 700603 and BAA-2146, were used as control strains.

2.2. Antibiotic resistance profile and determination of MDR strains by MAR index

The antibiotic susceptibility test against antibiotics belonging to different classes was performed using the Kirby Bauer disc diffusion method and microtiter broth dilution assay. The resistance profile of the clinical strains was evaluated against antibiotics (Oxoid, UK) including ampicillin (10µg), amoxicillin (10µg), gentamicin (10µg), amikacin (30µg), tetracycline (30µg), doxycycline (30µg), cefixime (5µg), cefepime (30µg), cefopodixime (10µg), ceftazidime (30µg), ceftriaxone (30µg), piperacillin/tazobactam (100/10µg), aztreonam (30µg), ciprofloxacin (5µg), imipenem (10µg), ertapenem (10µg), and meropenem (10µg). The clinical isolates were classified as susceptible, intermediate, and resistant using the CLSI guidelines. Whereas, colistin (the breakpoints <8 mg/L for susceptible and > 8 for resistant), and tigecycline (the breakpoints are <2 mg/L for Susceptible, 4 mg/L for intermediate, and > 8MIC for resistant) were determined by microbroth dilution using the EUCAST guideline. The Multiple Antibiotic Resistance (MAR) index was calculated by the number of drugs to which the isolate was resistant and divided by the total number of drugs tested. In this study, 0.21 or greater was considered as MDR strains, as resistant against more than three classes of drugs (Medugu et al., 2023).

2.3. ESBL phenotypic confirmatory test

A double disk synergy test was performed to detect third-generation cephalosporins ESBL *K. pneumoniae* clinical isolates, where ATCC 700603 was used as a positive control. A bacterial lawn was prepared on Mueller-Hinton Agar plates. A disc of Amoxicillin-clavulanate was placed in the centre of the MHA plate. The discs of cefepime, ceftazidime, cefixime, and cefopodixime were placed 1.5 -2 cm apart from each other. The next day, a positive result was noted as a keyhole or champagne-cork shape on plate (Kaur et al., 2013).

2.4. Phenotypic carbapenemase detection

The clinical isolates were streaked on ChromAgar™ as per the manufacturer's guidelines (mSuperCARBA™, France). The plate was incubated at 18-24 h at 37 °C, and the next day noted the metallic blue color was noted as positive carbapenemase production. Where BAA-2146 (ATCC) was used as a positive carbapenemase control (Moran Gilad et al., 2011).

2.5. Biofilm formation

The biofilm-forming potential of clinical isolates was determined using the crystal violet staining method²⁴. A hemolysis production was assessed by spotting bacterial suspension on rabbit blood agar. The hemolysis was noticed as alpha (green zone), beta (clear zone was seen), and gamma hemolysis (no zone). A positive string test was detected by stretching wire loop on the isolated colony, the string exhibited greater than 5 mm was identified as hypermucoviscous phenotype (Ahmed et al., 2016; Pereira and Vanetti, 2015).

2.6. Genotypic characterization of virulence factors and antibiotic resistance

Genomic DNA of clinical isolates of *K. pneumoniae* was isolated by

boiling method (Golpasand et al., 2024). The conventional PCR was performed for amplifying genes for biofilm formation (*FimH*, *MrkA*, *MrkD*) urease production (*Ure*), capsular gene (*Wzi*), efflux pump (*Acr*, *Tol*), outer membrane proteins (*OmpK35*, *OmpK36*), ESBL resistant genes (*bla_{SHVA}*, *bla_{SHVB}*, *bla_{TEM}*), and carbapenem resistant genes (*bla_{NDM}*, *bla_{OXA-48}*). The list of primers with annealing temperatures were mentioned in Table 1.

2.7. ERIC PCR for genetic relatedness

The PCR conditions for Enterobacterial Repetitive Intergenic Consensus Polymerase Chain Reaction (ERIC-PCR) were 94 °C for 3 min, 35 cycles of 1 min of denaturation at 94 °C, 2 min of annealing at 43.5 °C, 2 min of extension at 72 °C, with a final elongation time of 5 min at 72 °C. The ERIC-PCR product was electrophoresed in 1.5 % (w/v) agarose gels at 70 V using 1 × TBE buffer for 2 h. The amplified products were visualized on UV transilluminator. Genotypic variations were analyzed using the GelJ software (version 2.0). Unweighted Pair Group Method with Arithmetic Mean (UPGMA) cluster analysis was used to produce a dendrogram. The dendrogram was analyzed by Simpson's diversity index (SDI), using the following formula (Lagha et al., 2021).

$$SDI = 1 - (\sum n(n-1) / N(N-1))$$

2.8. Pearson coefficient correlation

To determine the correlation between antibiotic resistance and virulence factors, Pearson correlation coefficients (r) were calculated. The correlation map was generated using Hi-plot. The map determines the relationship between multidrug resistant, ESBLs, and carbapenemase strains with other virulence factors phenotypically and genotypically (Hao et al., 2022).

2.9. Statistical analysis

The data was analyzed through one-way ANOVA and Pearson coefficients (r) using GraphPad (version 8.0.1.244), and $p \leq 0.05$ values were considered statistically significant.

3. Results

3.1. Detection of ESBL, carbapenemases, and phenotypic resistance profile of clinical isolates of *Klebsiella pneumoniae*

A double disk synergy test was performed for the extended spectrum beta lactamase (ESBL) resistant strains (Fig. 1A). The clinical isolates were also cultured on ChromAgar media™ to validate the carbapenem resistance in *K. pneumoniae* strains (Fig. 1B). The Antibiotic susceptibility pattern showed the highest resistance against the penicillin class, i. e., ampicillin 98 % and amoxicillin 96.8 %. Class Cephalosporin showed resistance in cefepime 75 %, ceftazidime 75.7 %, ceftriaxone 78 %, cefixime 88 %, and cefopodixime 87.5 %. The beta-lactam combination of piperacillin/tazobactam depicted resistance in 54 % of isolates, whereas 75 % of the strains that were resistant to aztreonam belonged to the class monobactam. In the protein synthesis inhibitor class, strains showed resistance against tetracycline (46.0 %) and doxycycline (31.8 %). In class, aminoglycoside resistance against gentamicin (57 %) and amikacin (45 %) was observed. Ciprofloxacin, a fluoroquinolone, was found ineffective against 67.1 % of strains. Resistance against ertapenem (45 %), imipenem (44.5 %), and meropenem (73 %) confirmed the presence of carbapenem-resistant strains. The 41.7 % and 29.9 % strains were found resistant to last resort antibiotics, i.e., colistin and tigecycline, respectively (Fig. 1C). The MAR index was evaluated through antibiotic susceptibility data. It revealed that 90.2 % (102/113) of isolates were multidrug resistant (MDR), and ranged between 0.21 and 1 score (Fig. 1D). The multidrug-resistant *K. pneumoniae* strains were categorized into 16 clades with 72 antibiotic susceptibility patterns

Table 1
List of primers used in this study.

Primers	Primer Sequence	Product size (bp)	Annealing Temp (°C)
<i>bla_{TEM}</i>	F: 5'-TGCGGTATTATCCCGTGTG-3' R: 5'-TCGTCGTTTGGTATGGCTTC-3'	297	51 °C
<i>bla_{SHV-A}</i>	F: 5'-TCAGCGAAAAACACCTTG-3' R: 5'-TCCCGCAGATAAATCACC-3'	472	51 °C
<i>bla_{SHV-B}</i>	F: 5'-GGTTATGCGTTATATTCGCC-3' R: 5'-TTAGCGTTGCCAGTGCTC-3'	865	53 °C
<i>bla_{OXA-48}</i>	F: 5'-GCGTGTTAAGGATGAACAC 3' R: 5'-CATCAAGTTCAACCAACCG 3'	438	52 °C
<i>bla_{NDM}</i>	F: 5'-GTTTGGCGATCTGGTTTTC-3' R: 5'-GGAATGGCTCATCAGATC-3'	621	58 °C
<i>Acr</i>	F: 5'-ATCAGCGCGCGGATTGGTAAA-3' R: 5'-CGGGTTCGGGAAAATAGCGCG-3'	312	61 °C
<i>Tol</i>	F: 5'-ATCAGCAACCCGATCTGCGT-3' R: 5'-CCGGTGACTTGACGCAGTCCT-3'	527	59 °C
<i>Fim H</i>	F: 5'-ATGAACGCCTGGTCTTTGC-3' R: 5'-GCTGAACGCCTATCCCCTGC-3'	688	59 °C
<i>Mrk-A</i>	F: 5'-CGGTAAAGTTACCGACGTATCTTGACTG-3' R: 5'-GCTGTAAACCAACCGGTGTAAC-3'	475	60 °C
<i>MrkD</i>	F: 5'-AAGCTATCGCTGTACTTCCGCA-3' R: 5'-GGCGTTGGCGCTCAGATAGG-3'	340	53. °C
<i>OmpK36</i>	F: 5'-GAAATTTATAACAAAGACGGC-3' R: 5'-GACGTTACGTCGTATACTACG-3'	305	53 °C
<i>OmpK35</i>	F: 5'-CTCCAGCTCTAACCGTAGCG-3' R: 5'-GGTCTGTACGTAGCCGATGG-3'	241	51 °C
<i>Ure</i>	F: 5'-GCTGACTTAAGAGAAGCTTATG-3' R: 5'-GATCATGGCGCTACCTCA-3'	337	60 °C
<i>CAP(Wzi)</i>	F: 5'-GTGCCGCGAGCGCTTTCTATCTTGGTATCC3' R: 5'-GAGAGCCACTGGTCCAGAACTTCAACCGC-3'	581	55 °C
<i>rpo B</i>	F: 5'-GGCGAAATGGCGGAGAACCA-3' R: 5'-GAGTCTTCGAAGTTGTAACC -3'	1076 bp	55 °C
<i>ERIC</i>	F: 5'-ATGTAAGCTCCTGGGGATTAC-3' R: 5'- AAGTAAGTACTGGGGTGAGCG-3'		

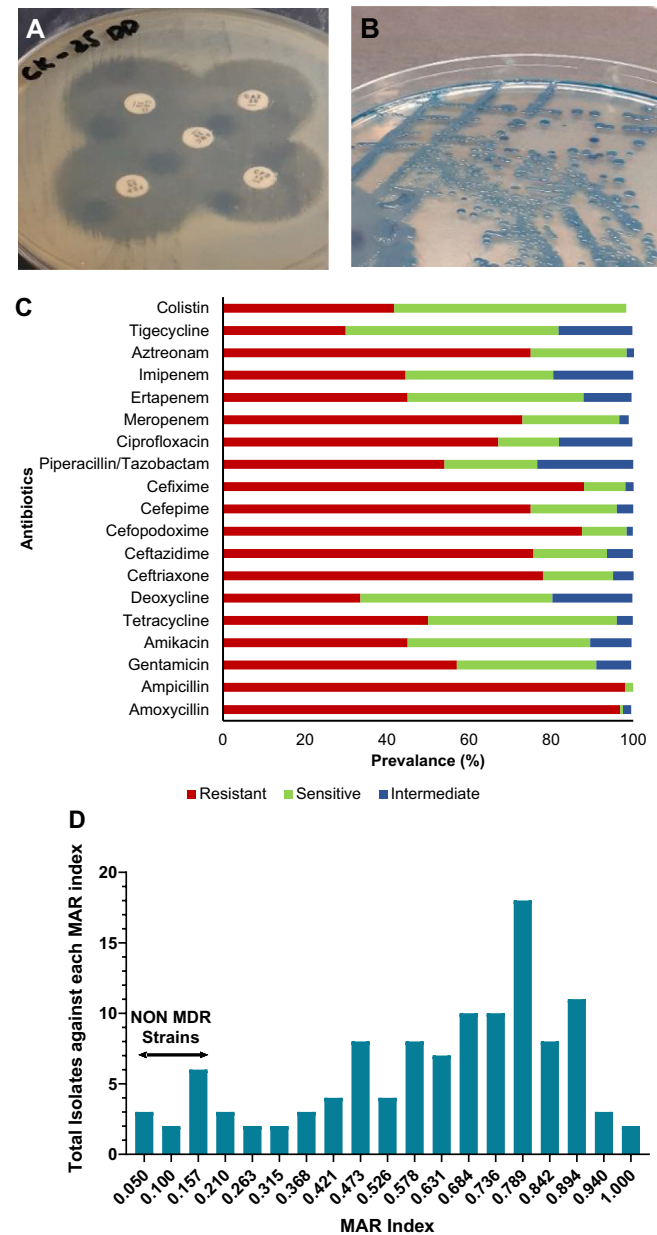


Fig. 1. Detection of ESBL and Carbapenemases Strains and Phenotypic Resistance Profile of *K. pneumoniae* Clinical Isolates. (A) Positive Double Disk synergy test for ESBL, (B) Blue metallic appearance on ChromAgar for Carbapenem positive strains (C) Antibiotic susceptibility profiles representing Resistant, Intermediate, and Susceptible in percent, (D) Prevalence of Non-MDR and MDR *K. pneumoniae* clinical isolates calculated by MAR index. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(Table 2).

3.2. Phenotypic virulence characterization and correlation with antibiotic resistance pattern

Blood agar was used for hemolysis test. Out of 113 isolates, 9.73 % and 5.30 % strains exhibited alpha and beta hemolysis, respectively (Fig. 2A). For hypermucoviscosity, 5.30 % strains showed positive string test (Fig. 2B). Overall, 97.3 % of strains were formed biofilm, among them 28.3 %, 24.77 %, and 44.2 % strains produced strong, moderate or weak biofilm respectively (Fig. 2C).

The phenotypic characterization of hypermucoidity, hemolysis, and

Table 2
Phenotypic Resistance Profile of Clinical Isolates of *K. pneumoniae*.

S. NO	MAR Index	Resistant isolates N (%)	Phenotypic Resistance Profile (No. of Isolates)
1	0.21	3 (2.9 %)	CFM, AMP, MEM, CN
2	0.263	2 (1.9 %)	CFM, AML, AMP, CN, MEM
3	0.315	2 (1.9 %)	CFM, AML, CIP, IMI, ERT, CT
4	0.368	3 (2.9 %)	CFM, AML, CPD, ERT, MEM, CT CFM, AML, AMP, MEM, CIP, AK, CN CFM, AML, AMP, CPD, ATM, CIP, MEM
5	0.421	4 (3.9 %)	CFM, AML, AMP, CPD, CIP, TET, DO CFM, AML, FEP, CRO, CAZ, CPD, AMP, MEM (02)
6	0.473	8 (7.9 %)	CFM, AML, PEP, AMP, CN, CIP, IMI, MEM CFM, AML, PEP, CPD, AMP, CIP, MEM, CT CFM, AML, CRO, CAZ, CPD, ATM, AMP, AK, MEM CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, TG CFM, AML, PEP, CPD, AMP, AK, IMI, ERT, MEM CFM, AML, FEP, CRO, CAZ, CPD, AMP, MEM, CT (02) CFM, AML, FEP, CRO, CPD, ATM, AMP, CN, TE CFM, AML, CAZ, CPD, AMP, ERT, MEM, CT, TG CFM, AML, CPD, AMP, CN, CIP, MEM, TE, CT CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, AMP, CIP, MEM, TG CFM, AML, PEP, CRO, CAZ, CPD, ATM, AMP, AK, TE CFM, AML, CPD, AMP, CN, CIP, MEM, TE, DO, CT, TG
7	0.526	4 (3.9 %)	CFM, AML, FEP, CRO, CAZ, CPD, AMP, CIP, MEM, TG CFM, AML, PEP, CRO, CAZ, CPD, ATM, AMP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, TE CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, TE.TG CFM, AML, FEP, CRO, CAZ, CPD, AMP, MEM, TE, MEM, CT CFM, AML, PEP, CPD, AMP, CN, CIP, MEM, TE, CT, TG
8	0.578	8 (7.9 %)	CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, MEM, TE CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, CT, TG CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, TE CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, TE.TG CFM, AML, FEP, CRO, CAZ, CPD, AMP, MEM, TE, MEM, CT CFM, AML, PEP, CPD, AMP, CN, CIP, MEM, TE, CT, TG
9	0.631	7 (6.9 %)	CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, TE, DO (02) CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, IMI, ERT, CT CFM, AML, PEP, FEP, CRO, CAZ, CPD, AMP, AK, IMI, ERT, MEM CFM, AML, PEP, CRO, CAZ, CPD, ATM, AMP, CN, MEM, CT CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, MEM, TE, DO, CT CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, MEM, CT, TG
10	0.684	10 (9.9 %)	CFM, AML, PEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, TE, DO CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, IMI, MEM, TE CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, TE, DO CFM, AML, PEP, FEP, CRO, CAZ, CPD, AMP, CN, AK, IMI, ERT, MEM CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, IMI, ERT, MEM CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, MEM, TE

(continued on next page)

Table 2 (continued)

S. NO	MAR Index	Resistant isolates N (%)	Phenotypic Resistance Profile (No. of Isolates)
11	0.736	10 (9.9 %)	CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, TE, DO
			CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, TE, CT, TG
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, TE, CT, TG
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, TE
			FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, ERT, MEM, TG
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, AK, CIP, MEM, TE, DO
			CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, MEM, TE, DO, TG
			CFM, AML, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, MEM, CT, TG
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, ERT, MEM (02)
			CFM, AML FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, IMI, ERT, MEM, TE
12	0.789	18 (17.9 %)	CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, MEM, TE, DO (04)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM (11)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, TE, DO, CT (01)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CIP, IMI, ERT, MEM, TE, DO (01)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, IMI, ERT, MEM, CT, TG (01)
			CFM, AML, PEP, FEP, CRO, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TG
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, IMI, ERT, TE, DO, TG
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO (05)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, ERT, MEM, TE, DO, CT, TG
13	0.842	8 (7.9 %)	CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, CIP, IMI, ERT, MEM, TE, DO (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT (02)
14	0.894	11 (10.9 %)	CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
15	0.94	3 (2.9 %)	CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
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			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
16	1	2 (1.9 %)	CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
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			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)
			CFM, AML, PEP, FEP, CRO, CAZ, CPD, ATM, AMP, CN, AK, CIP, IMI, ERT, MEM, TE, DO, CT, TG (02)

biofilm formation ranging from weak to strong virulence profile can also be correlated with the prevalence of emerging ESBLs and carbapenemase strains. All the alpha and beta hemolytic strains were ESBL-positive, and 88 % of them were also carbapenem-resistant strains. Similarly, all

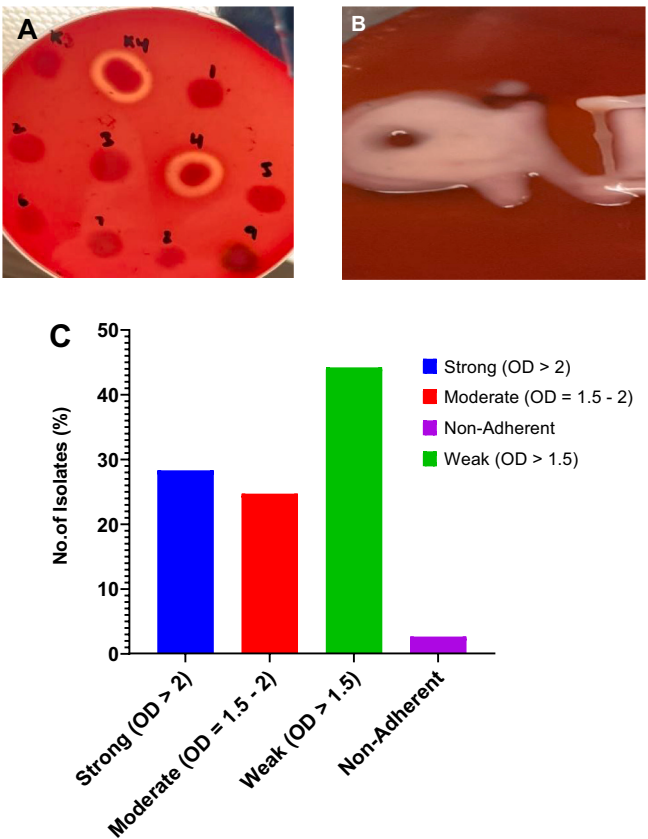


Fig. 2. Phenotypic Characterization of *K. pneumoniae* Clinical Isolates. (A): Hemolysis Assay, (B) String Test for hypermucoviscous Strains of *K. pneumoniae*, (C) Biofilm Forming Potential of Clinical strains of *K. pneumoniae*.

hypermucoid strains (5.3 %) were ESBL and carbapenemase positive. Moreover, all ESBLs *K. pneumoniae* isolates are biofilm forming, whereas 80 % of carbapenemase form weak, 60.7 % moderate, and 84.3 % formed strong biofilm formers (Supplementary table 1).

3.3. Resistance patterns based on source distribution

In total, 113 isolates were identified as *K. pneumoniae*, the strains were isolated from different sources which include urine (49.5 %), blood (32.7 %), pus swab (7.07 %), tracheal aspirate (4.42 %), pleural tissue (4.42 %) and cerebrospinal fluid (1.76 %) (Fig. 3A). On a gender basis, 68 % and 32 % of the strains were isolated from male and female patients, respectively. The prevalence of isolated antibiotic-resistant strains against carbapenem and ESBL was from urine, followed by blood, pus, tracheal aspirates, and pleural tissue. The high frequency of hypermucoid (HMV) strains was isolated from pleural tissue (60 %), and the lowest was from urine (3.5 %), followed by blood (2.7 %) isolates. The strains isolated from tracheal aspirate were strong biofilm formers (5 out of 6), while only 1 strain formed a moderate biofilm. The weak biofilm formers were isolated from urine (53.5 %), blood, pleural tissue, CSF, and pus (Fig. 3B).

3.4. Genotypic resistance profile of clinical isolates of *Klebsiella pneumoniae*

All clinical isolates of *K. pneumoniae* were confirmed by *16SrRNA* and *rpoB* gene. The capsular gene *CAP (Wzi)* was found among 55.4 % isolates. The PCR results showed the presence of urease (*Ure*) (61.06 %), outer membrane protein [*OmpK 35* = 49.50 %], (*OmpK 36* = 38.9 %)], efflux pump [*Tol* (53.09 %), *Acr* (40.7 %)], biofilm associated genes

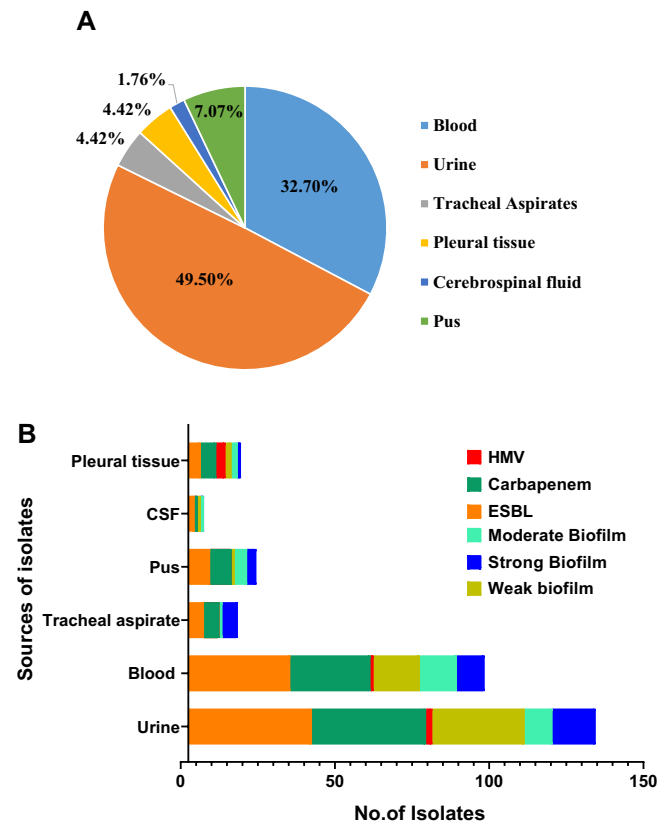


Fig. 3. Source distribution and virulence correlation of clinical isolates of *K. pneumoniae*. (A) Clinical origin of extracted isolates in percentage, (B) Source-wise prevalence of virulence factors, hypermucoidity (HMV), biofilm, and antibiotic resistance in ESBL and carbapenemase strains.

[*Mrk-A* (43.3 %), *Fim-H* (38.90 %), *Mrk-D* (64.6 %)], *ESBL* genes [*bla_{TEM}* (76.1 %), *bla_{SHV-A}* (25.6 %), *bla_{SHV-B}* (24.7 %)], and carbapenemase gene [*bla_{NDM}* (13.27 %), *bla_{OXA-48}* (15 %)] in clinical isolates as shown in Fig. 4.

3.5. Determination of genetic diversity by ERIC PCR

The fingerprinting patterns of clinical isolates of *K. pneumoniae* were generated by ERIC PCR. The maximum 13 bands were generated between 50 and 3000 bp, and the discriminatory index was calculated using Simpson's index of diversity. The highest clustering size was 15, which was obtained at 80 % similarity with a 0.67 discriminatory index.

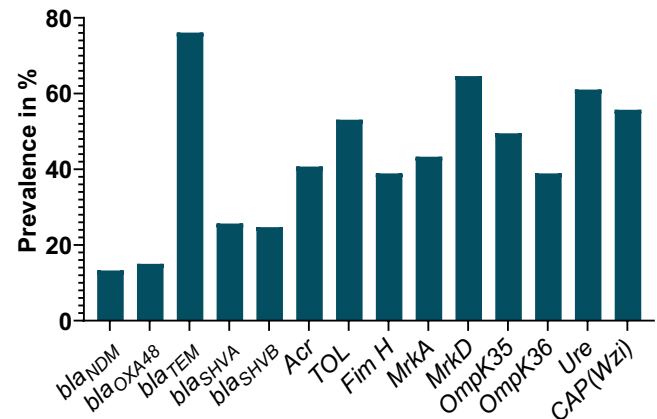


Fig. 4. Genotypic Resistance Profile of *K. pneumoniae* Clinical Isolates.

Moreover, the cutoff value was set at the lowest clustering at 71 % (Table 3 & Fig. 5 (See Dendrogram press Ctrl & Click here)). The clades (I-VII) were formed at 71 % (111 strains), representing diverted resistance patterns due to different antibiotic resistance genes (*bla_{SHVA}*, *bla_{SHV}*, *bla_{TEM}*, *bla_{NDM}*, *bla_{OXA-48}*), efflux pumps (*Acr*, *Tol*), outer membrane porins (*OmpK 35*, *OmpK 36*), and biofilm-forming potential (110/113). Two strains (9 and 2) were excluded from analysis. Clade III had acquired highest number of strains and obtained from different sources and acquired ESBLs, carbapenemase and other virulence genes (Table 4). Moreover, only four subclades G1-G4 from clade III formed at 100 % similarity, and each subclade found the same resistance pattern (Supplementary table 2).

3.6. Correlation analysis

Pearson's correlation coefficient analysis was used to assess the correlation between phenotypic and genotypic characteristics and other virulent factors of MDR strains of clinical *K. pneumoniae* (Fig. 6). All antibiotic resistance genes had a positive correlation with their phenotypes. Such as *bla_{TEM}* ($r = 0.96$, $p = 1.8 \times 10^{-11}$), *bla_{SHVA}* ($r = 0.84$, $p = 2.84 \times 10^{-6}$) and *bla_{SHVB}* ($r = 0.77$, $p = 7 \times 10^{-5}$) showed positive correlation with cephalosporins [(ESBLs(P)]. Similarly, BF strains also showed a strong correlation with biofilm-associated genes *FimH* ($r = 0.85$, $p = 2.09 \times 10^{-6}$), *MrkA* ($r = 0.86$, $p = 6.4 \times 10^{-7}$), and *MrkD* ($r = 0.92$, $p = 7.9 \times 10^{-9}$). The heatmap also displayed that MDR clinical strains were positively correlated with biofilm-forming strains ($r = 0.99$, $p = 4.4 \times 10^{-16}$), and similarly with efflux pump ($r = 0.94$, $p = 5.5 \times 10^{-10}$), and outer membrane porins ($r = 0.88$, $p = 1.4 \times 10^{-6}$).

4. Discussion

Klebsiella pneumoniae is a notorious pathogen and is involved in hospital and community acquired infections. It causes life-threatening infections namely pneumonia, urinary tract infection, septicemia, meningitis and liver abscess. Similarly, in this research, *K. pneumoniae* ($n = 113$) were isolated from urine (49.5 %), blood (32.7 %), pus (7.07 %), pleural (4.42 %), tracheal (4.42 %), and cerebrospinal fluid (1.76 %), suggesting their involvement in causing a broad spectrum of infections in local clinical settings(Palmeiro et al., 2019; Xianguang et al., 2016).

Globally, the prevalence of nosocomial infections is rapidly increasing due to the extensive use of antibiotics which led to multidrug resistant (MDR) *K. pneumoniae* strains. In this study, 90.2 % of *K. pneumoniae* strains found as multidrug resistant. Similar studies from other researcher found the prevalence of MDR *K. pneumoniae* between 83 % to 92 % in their samples(Lagha et al., 2021; Mirzaie and Ranjbar, 2021).It is observed that the MDR strains of *K. pneumoniae* evolved from extended spectrum beta lactamases (ESBL) to carbapenemase(Hosdurg Bhaskar et al., 2019). Therefore, the presence of ESBL resistance in our samples was identified, and the results indicated that 85 % ($n = 97$) of *K. pneumoniae* strains were resistant to third-generation cephalosporins (ESBLs). The most prevalent ESBL genes in isolated strains were *bla_{TEM}*, *bla_{SHV-A}*, and *bla_{SHV-B}*. These results are similar to other studies that found *bla_{TEM}* as the most prevalent gene, followed by the *bla_{SHV}* gene (Kang et al., 2020; Hussein et al., 2023). In terms of correlation with other virulent factors, it was observed that *K. pneumoniae* isolates

Table 3
Discriminatory index of *K. pneumoniae* Isolates based on dendrogram generated by ERIC PCR.

Discriminatory Index	Cluster Size	No. of Clusters	Similarity %
0.85	5,2,3,2,4,2,5,6,2,23,13,3,2,2	14	90 %
0.67	7,3,2,2,4,3,2,6,2,59,2,4,4,2,3	15	80 %
0.48	11,2,78,2,11,3,4	7	71 %

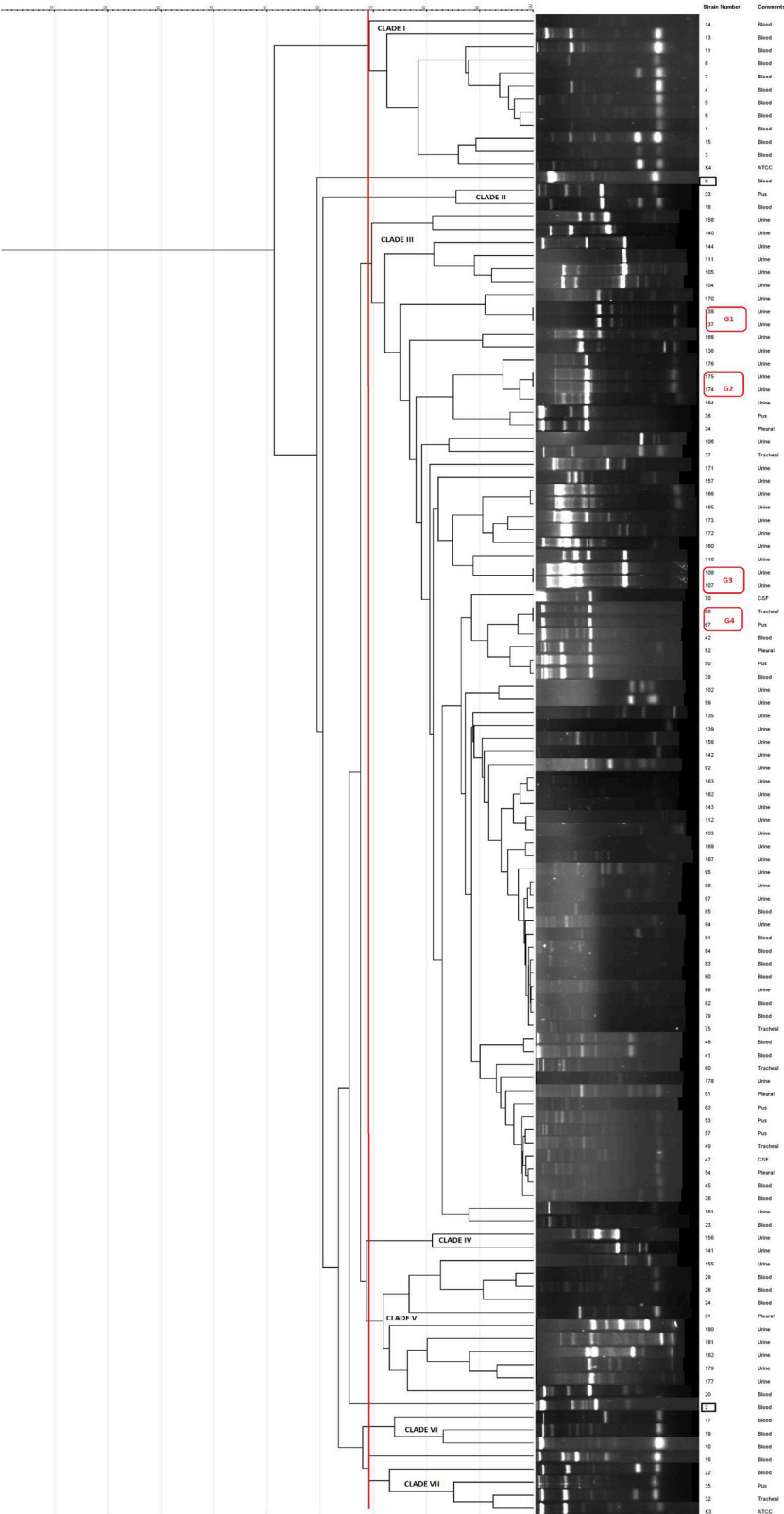


Fig. 5. Genetic similarity among *K. pneumoniae* Clinical isolates determined by constructing dendrogram using Gel J (Version 2) software. ((See Dendrogram press Ctrl & Click here).

producing ESBLs also acquired hypermucoidity and have strong biofilm-forming capabilities, which further decreases the treatment options and enhances the pathogenicity of strains (Supplementary table 1). Then, carbapenems are considered as priority antibiotic therapy for the

infections caused by ESBLs in *K. pneumoniae*. However, carbapenem-resistant *K. pneumoniae* (CRKP) has evolved and is involved in high morbidity and mortality. The prevalence of carbapenemase strains is emerging worldwide; therefore, the World Health Organization (WHO)

Table 4

Diversity of *K. pneumoniae* Isolates based on clade-wise resistance pattern established by ERIC PCR.

Clades	Total Strains	Source	Resistant genes (No. of Isolates)	Biofilm Potential
CLADE I	11	Blood	<i>bla_{SHVA}</i> (3), <i>bla_{SHVB}</i> (11), <i>bla_{TEM}</i> (3), <i>Acr</i> (9) <i>Tol</i> (11), <i>OmpK35</i> (4), <i>OmpK65</i> (1)	Weak, Moderate
CLADE II	2	Pus, Blood	<i>bla_{SHVB}</i> (1), <i>Acr</i> (2), <i>Tol</i> (2), <i>OmpK35</i> (1), <i>OmpK36</i> (1)	Weak, Strong
CLADE III	78	Urine (46), Pus (5), Pleural (3), Tracheal (5), CSF (2), Blood (17)	<i>bla_{SHVA}</i> (24), <i>bla_{SHV}</i> (17), <i>bla_{TEM}</i> (62), <i>bla_{NDM}</i> (13), <i>bla_{OXA-48}</i> (15), <i>Acr</i> (27), <i>Tol</i> (34), <i>OmpK35</i> (38), <i>OmpK36</i> (34), <i>bla_{TEM}</i> (2), <i>Tol</i> (1)	Non-adherent, Weak, Moderate, Strong
CLADE IV	2	Urine	<i>bla_{SHVA}</i> (1), <i>bla_{SHVB}</i> (3), <i>bla_{TEM}</i> (6), <i>bla_{NDM}</i> (2), <i>bla_{OXA-48}</i> (2), <i>Acr</i> (5), <i>Tol</i> (6), <i>OmpK35</i> (6), <i>OmpK36</i> (8)	Weak, Moderate
CLADE V	11	Blood (5), Urine (5), Pleural (1)	<i>bla_{SHVA}</i> (1), <i>bla_{SHVB}</i> (1), <i>bla_{TEM}</i> (1), <i>Acr</i> (2), <i>Tol</i> (3), <i>OmpK35</i> (3), <i>OmpK36</i> (2)	Non-adherent, Weak, Moderate, Strong
CLADE VI	3	Blood	<i>bla_{SHVA}</i> (1), <i>bla_{SHVB}</i> (1), <i>bla_{TEM}</i> (1), <i>Acr</i> (2), <i>Tol</i> (3), <i>OmpK35</i> (3), <i>OmpK36</i> (2)	Weak, Moderate
CLADE VII	4	Blood (2), Tracheal (1), Pus (1)	<i>bla_{SHVA}</i> (2), <i>bla_{SHVB}</i> (1), <i>bla_{TEM}</i> (3), <i>bla_{OXA-48}</i> (1), <i>Acr</i> (3), <i>Tol</i> (3), <i>OmpK35</i> (3), <i>OmpK36</i> (3)	Weak, Strong

has listed *CRKP* as a priority pathogen. In China, the rate of carbapenemase strains has surged up to 11.88 in the last 15 years(Tian et al., 2023). Moreover, meta-analysis also revealed the prevalence of carbapenem-resistant *K. pneumoniae* strains colonizes between 0.13 and

22 % (Karampatakis et al., 2023). Our data revealed 73 % of the isolates were *CRKP* (meropenem), similar to recent studies, which reported 82 % and 96.3 % *CRKP* isolates(Imtiaz et al., 2021; Zahedi Bialvaei et al., 2023). Furthermore, the plasmid-associated epidemic *CRKP* genes (*bla_{OXA-48}*, *bla_{NDM}*) were also studied, and their prevalence was found as 15 % and 13.27 %, respectively. The prevalence *bla_{NDM}* is alarming compared to other reported data, where *bla_{NDM}* strains were reported as superbugs, and prevalence was found as 16.5–39 % (Holt et al., 2015; Hosdurg Bhaskar et al., 2019; Kot et al., 2023). This study also revealed that carbapenemase hydrolyzes penicillin, monobactams, cephalosporin, aminoglycosides, and fluoroquinolones, which is according to other studies. The disruption of outer membrane porins is also involved in the resistance mechanism of antimicrobials by decreasing the permeability of outer membrane to drugs(Ferreira et al., 2019; Tahae et al., 2023). This research also displayed the role of outer membrane porins genes (*OmpK35* and *OmpK36*) ($r = 0.90$, $r = 0.83$, $p < 0.0001$ respectively) that confers the carbapenem resistance.

Another contributing factors in resistance mechanism is the activation of efflux pump. *AcrAB-TolC* is well studied efflux pump system in *K. pneumoniae* and also found in carbapenem resistance. In this study, the prevalence of efflux pump genes *Tol* and *Acr* were found to be 53.09 % and 40.7 %, respectively. The efflux pump genes also showed positive correlation with ESBLs and carbapenem [CARBs(P)] (Fig. 7). These resistance trends have been reported in clinical settings, thus, health organizations proclaim this an alarming situation(Imtiaz et al., 2021; Zahedi Bialvaei et al., 2023). Hence, colistin and tigecycline are the final line of treatments for *CRKP* strains. However, resistance against these ultimate treatments has also been reported in Pakistan(Furqan et al., 2022). Notably, our study also found strains that are resistant to colistin (41.7 %) and tigecycline (29.9 %), which is very well correlated with previous studies(Furqan et al., 2022; Hameed et al., 2021). This emerging resistance to last resort drugs showed the presence of the *mcr* gene and modified LPS structures in *K. pneumoniae* strains. Cumulatively, the phenotypic resistance profile of *K. pneumoniae* isolates highlighted that highly variant strains are circulating in our population, with

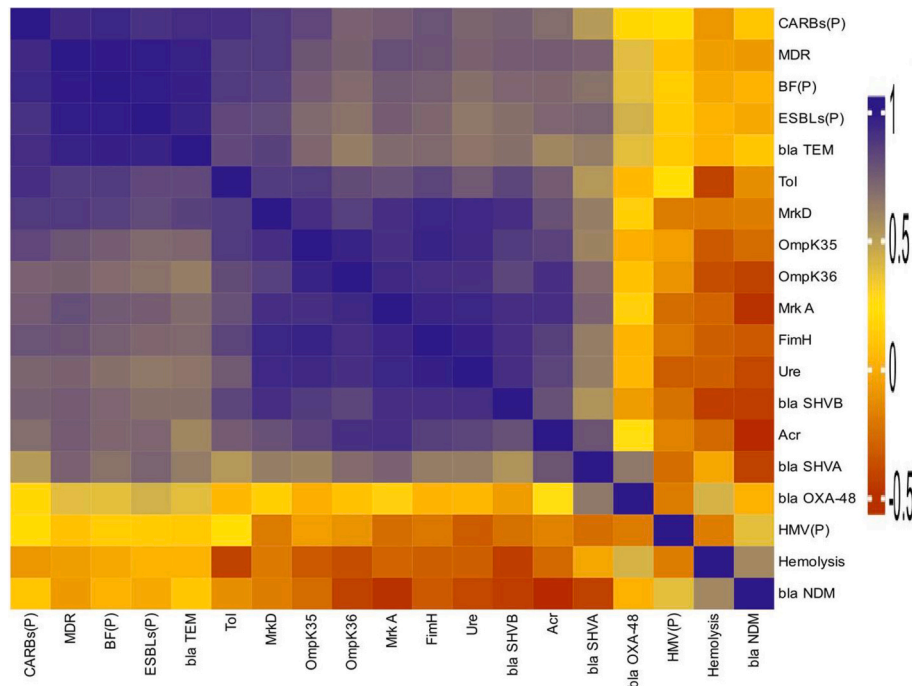


Fig. 6. Correlation Analysis. The heat map was generated using Hi-plot by applying the Pearson correlation coefficient. Blue is positive, and yellow is moderate correlation, while brown is negative; however, the color intensity indicates the degree of correlation between different factors. Abbreviations: Carbapenem phenotypes [CARBs (P)], Biofilm forming phenotypes [BF(P)], Hypermucoviscous Phenotypes [HMV(P)]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

72 (16 clades) different antibiotic resistance patterns.

The correlation between the biofilm-forming capability and their genes encoding fimbriae (*FimH* and *MrkD*) with drug resistance was also investigated in clinical isolates. The result showed that 97.3 % of isolates are biofilm-forming, which is a major factor that hinders drug delivery and facilitates infection. The clinical isolates of *K. pneumoniae* can form biofilms, ranging from weak to strong. Surprisingly, the same correlation was observed in another study (Mirzaie and Ranjbar, 2021). *K. pneumoniae* is characterized by type 1 and 3 fimbriae as a pathogenic factor. The fimbriae genes involved in biofilm formation include *FimH* (Type 1 fimbriae), *MrkA*, and *MrkD* (Type 3 fimbriae), and their prevalence in clinical isolates was found to be 38.9 %, 43.3 %, and 64.6 %, respectively (Kot et al., 2023). It was also observed that biofilm-forming strains had acquired resistance against ESBLs and carbapenemase *K. pneumoniae* strains (Table 3). Hypermucoidity is also a virulent trait of *K. pneumoniae* that targets immunocompetent populations. Our findings showed that 5.3 % of the clinical isolates of *K. pneumoniae* had the hypermucoviscous (HMV) phenotype. Furthermore, the *Wzi* (CAP) gene is a conserved gene in all 78 different capsular serotypes. Almost 55.4 % of the isolates showed the presence of the conserved capsular gene *Wzi* (CAP) in this study. Surprisingly, the presence of K1 and K2 serotypes was not found in this study, which was more prevalent capsular serotypes among nosocomial isolates reported in other studies (Brisse et al., 2013; Visciano et al., 2018). This might be due to the genetic diversity, or methodology limitation. The genetic variability is one of the key features to found diversity within the population. Therefore, the genetic similarities between MDR *K. pneumoniae* isolates was performed through ERIC PCR (Lagha et al., 2021; Visciano et al., 2018; Zehra et al., 2025). It analyzed that the higher the discriminatory value, the reduced the diversity within the population (Bobbadi et al., 2023; Kundu et al., 2022). In this study, highest discriminatory index was found as 0.85 at 90 %, and 0.48 at 71 %. Thus, at 71 % (cut off) the genetic diversity was observed, forming 7 clades. These clades indicate heterogeneous strains are circulating in our clinical setup and strains are genetically diverse due to resistance pattern of antibiotic resistant genes, involvement of other virulent factors such as biofilm (Table 4).

In order to determine the prevalence of ESBL and CRKP strains and their association with other pathogenic factors, Pearson correlation study was performed. The Correlation analysis shows a strong relationship between multi-resistant strains with virulence factors. The ESBLs ($r = 0.98$) and carbapenem resistant Strains [CARBs(P)] ($r = 0.96$) positively correlated to biofilm (BF) ($p < 0.0001$). Similarly, efflux pump gene ($r = 0.92$) outer membrane porins ($r = 0.88$) also showed strong correlation with multidrug-resistant strains ($p < 0.0001$). While the hypermucoviscous (HMV) phenotype is positively linked to antibiotic resistance, which is consistent with the other literature (Zhan et al., 2017).

This study provides an overview of the antibiotic resistance and virulence scenario of the *K. pneumoniae* strains in our healthcare system. The presence of highly resistant pathogens demands the need for antibiotic stewardship, proper AMR surveillance, and infectious control measures to deal with emerging AMR superbugs.

5. Conclusion

The study revealed a significant prevalence of extended spectrum beta lactam and carbapenem-resistant *K. pneumoniae* isolates in our population. The treatment of illness associated with carbapenem-resistant *K. pneumoniae* is hampered by its ability to form biofilm, its efflux pump, outer membrane porins, and other virulence factors such as capsule formation and hypermucoviscous characteristics. Furthermore, the high genetic similarity and diverse genetic differences in hospital settings suggest a clonal dissemination of *K. pneumoniae* strains. The study calls for urgent action on antimicrobial stewardship programs in a country to reduce the threat of rising CRKP-associated infections.

6. Limitation

The presented study fails to discriminate between K1 and K2 Capsular typing, which might be due to the absence of these capsular types, sample variation, and method limitations. Due to the budget constraint, the whole genome sequence cannot be a part of this study. Next, the samples will be collected from different districts and regions of Karachi to give a better prevalence rate and AMR in clinical *K. pneumoniae* isolates.

The ethical approval and consent to participate

Not Applicable.

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CRediT authorship contribution statement

Rabia Ilyas: Writing – original draft, Project administration, Methodology, Investigation. **Sidrah Asghar:** Project administration, Methodology, Investigation. **Moatter Zehra:** Project administration, Methodology, Investigation. **Yamina Usmani:** Methodology, Investigation. **Rao Muhammad Abid Khan:** Resources, Investigation. **Zulfiqar Ali Mirani:** Resources, Methodology. **Syed Abid Ali:** Writing – review & editing, Investigation. **Mohammad Y. Alshahrani:** Writing – review & editing, Resources, Funding acquisition. **Ajmal Khan:** Writing – review & editing, Resources. **Ahmed Al-Harrasi:** Writing – review & editing, Resources. **Ayaz Ahmed:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2025.105785>.

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